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Pimpri Chinchwad College of Engineering

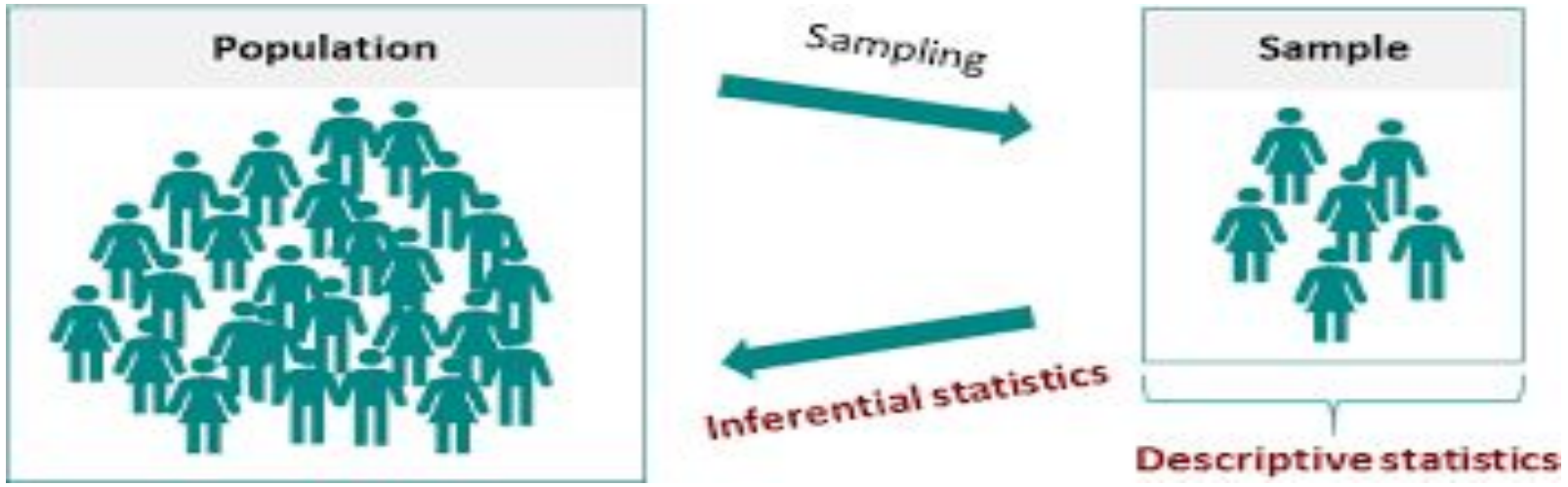
Department of Applied Sciences and Humanities

Topic: Sampling Techniques and Hypothesis Testing

Content

- Inferential statistics
- Introduction of Sampling
- Purpose of Sampling
- Types of Sampling
- Sample Size
- Confidence interval and Margin error

Inferential statistics



Inferential statistics use a random sample to draw conclusions about the population. Typically, it is not practical to obtain data from every member of a population. Instead, we collect a random sample from a small proportion of the population. From the sample, statistical procedures can infer the likely properties of the population.

Statistical Analysis

- Descriptive Statistics (Analysis)
  - Graphical : Organizing and presenting the data (eg. box plots, probability plots)
  - Numerical: Summarizing the sample set (eg. mean, mode, range, variance, moments)
- Inferential
  - Estimation: Estimate parameters of the pdf along with its confidence region
  - Hypotheses testing: Making judgements about f(x) and its parameters

Need for sampling

- PDFs of RVs establishes theoretical framework. But
  - Entire sample space may not be known
  - Parameters of distribution may not be known
- From a finite sample derive conclusions about the pdf and its parameters
- Sample (or observation) set is assumed to be sufficiently representative of the entire sample space
  - Proper sampling procedures and design of experiments to be used for obtaining the sample

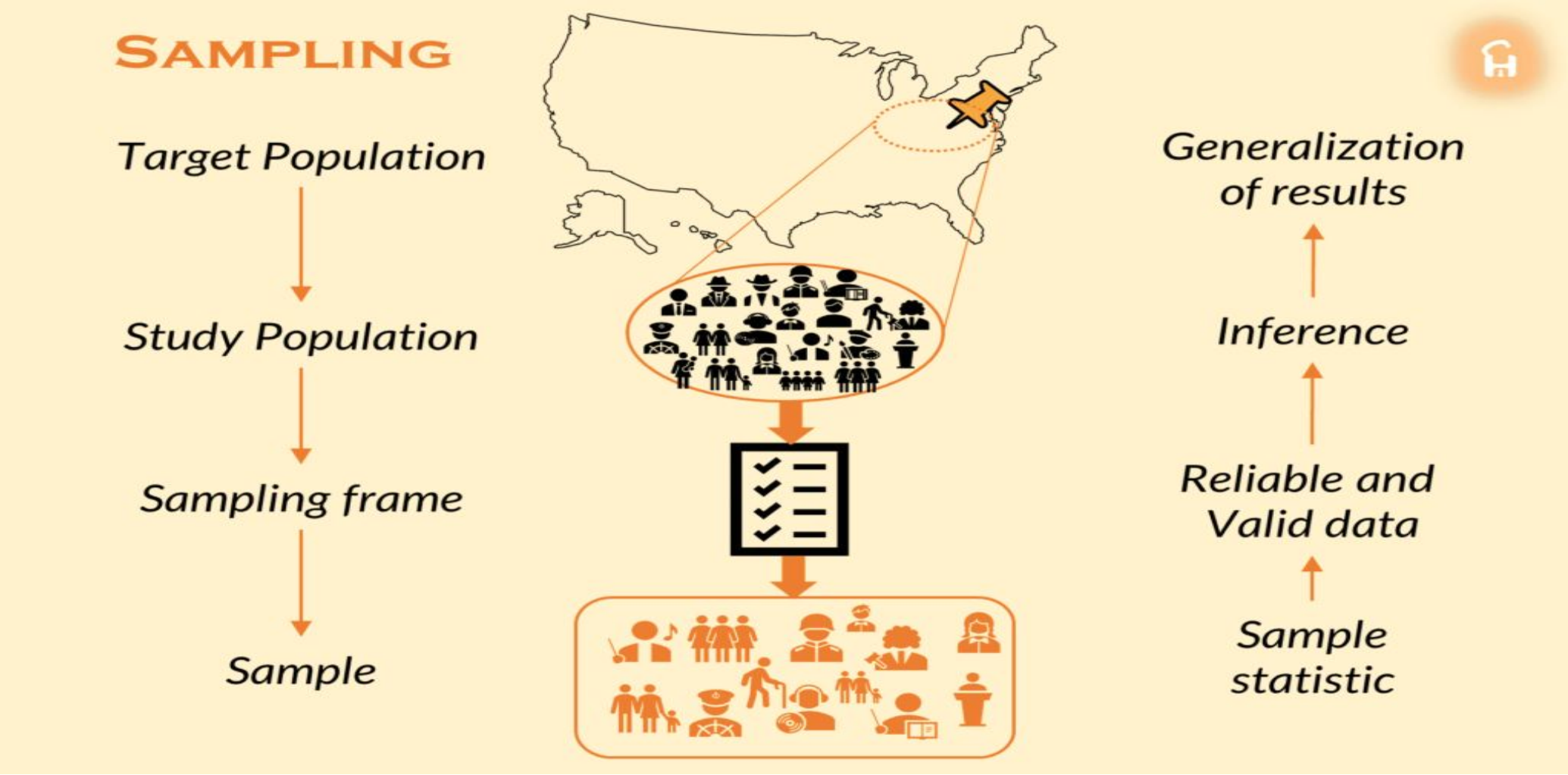
Basic Concepts

- Population: Set of all possible outcomes of a random experiment characterized by f(x)
- Sample set (realization) : Finite set of observations obtained through an experiment
- Inference: Conclusion derived regarding the population (pdf, parameters) from the sample set
  - Inference made from a sample set is also uncertain since it depends on the sample set which is one of many possible realizations

Why sampling?

- The sampling costs are cheaper and more practical than reaching every member of a target population. It saves you money and time.
- With sampling, you have a higher degree of control with less complexity.
- A sample can be more accurate as it eliminates exposure to non-sampling errors like non-response-bias
- It is applied when there is difficulty in having censuses.

Steps of Sampling Process



Types of Data and Data Collection

Primary data

- Personal investigation, Collection Via Investigators Questionnaires, Telephonic Investigation

Secondary data

- Official publications such as the Ministry of Finance, Statistical Departments of the government, Agricultural Statistical boards, etc. Semi-official sources include State Bank
- Data published by Chambers of Commerce and trade associations and boards.
- Articles in the newspaper, from journals and technical publications

Definition of Sampling

Sampling is a technique of selecting individual members or a subset of the population to make statistical inferences from them and estimate characteristics of the whole population.

What is Good Sample?

The sample must be:

1. representative of the population;
2. appropriately sized (the larger the better);
3. unbiased;
4. random (selections occur by chance);

Merits of Sampling

1. Size of population
2. Fund required for the study
3. Facilities
4. Time

Types of Sampling Methods



Probability Sampling

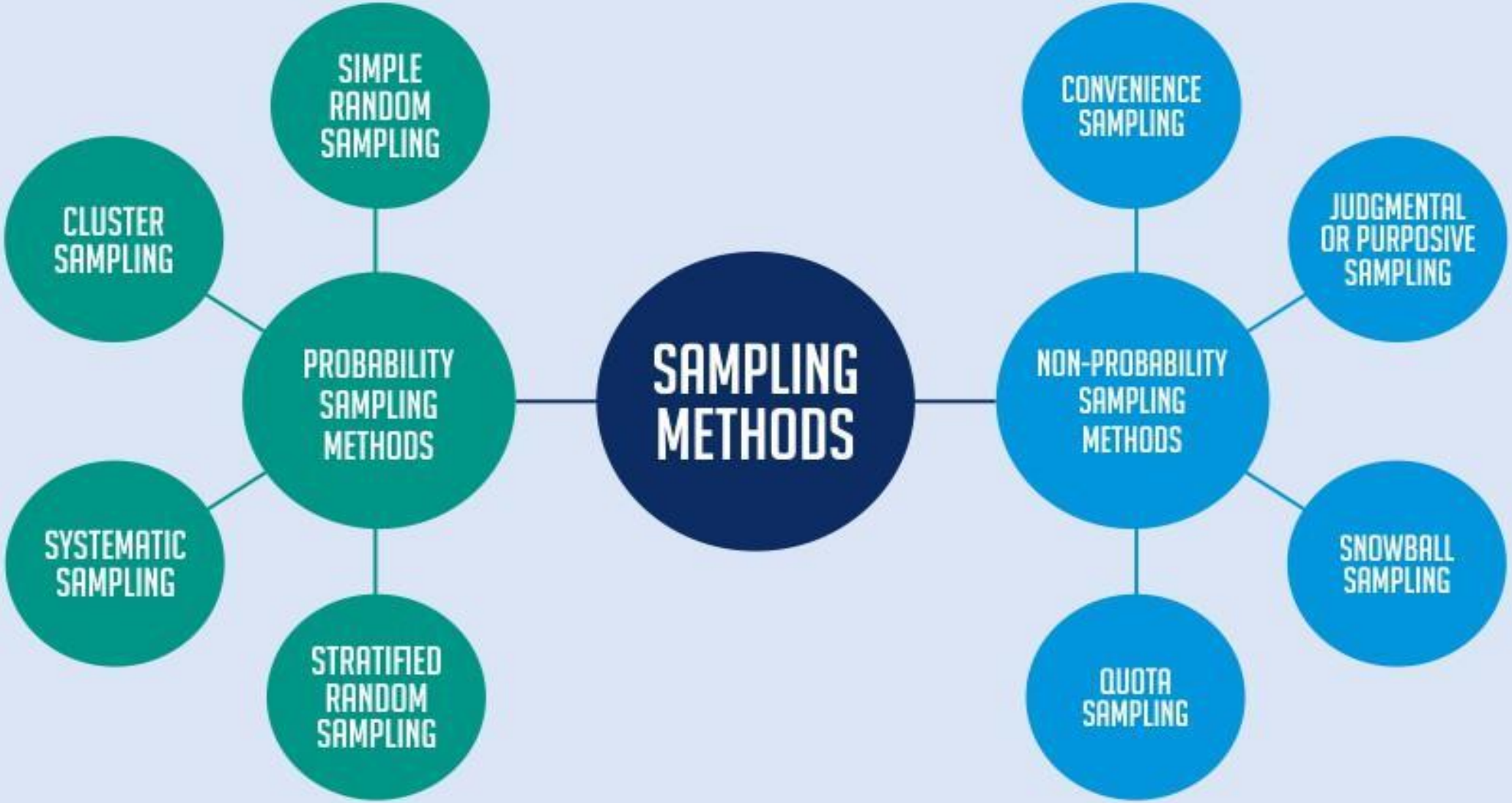
Probability sampling utilizes random sampling techniques to create a sample. This group of sampling methods give all the members of a population equal chances of being selected.

Non-probability Sampling

It is a group of sampling techniques where the samples are collected in a way that does not give all the units in the population equal chances of being selected. Probability sampling does not involve random selection at all.

Types of Non-probability

QuestionPro



Types of Probability Sampling

Random Sampling

When: There is a very large population and it is difficult to identify every member of the population.

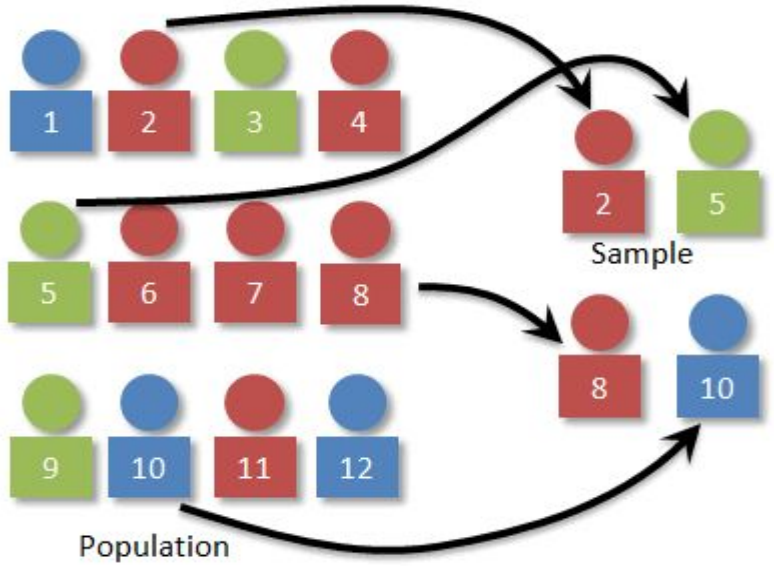
How: The entire process of sampling is done in a single step with each subject selected independently of the other members of the population

Pros: In this technique, each member of the population has an equal chance of being selected as subject.

Cons: When there are very large populations, it is often difficult to identify every member of the population and the pool of subjects becomes biased.

Types of Probability Sampling

Want to study and understand the rice consumption pattern across rural India? While it might not be possible to cover every household, you could draw meaningful insights by building your sample from different districts or villages (depending on the scope).



Types of Probability Sampling

Systematic Sampling

When: Your given population is logically homogenous.

How: In a systematic sample, after you decide the sample size, arrange the elements of the population in some order and select terms at regular intervals from the list.

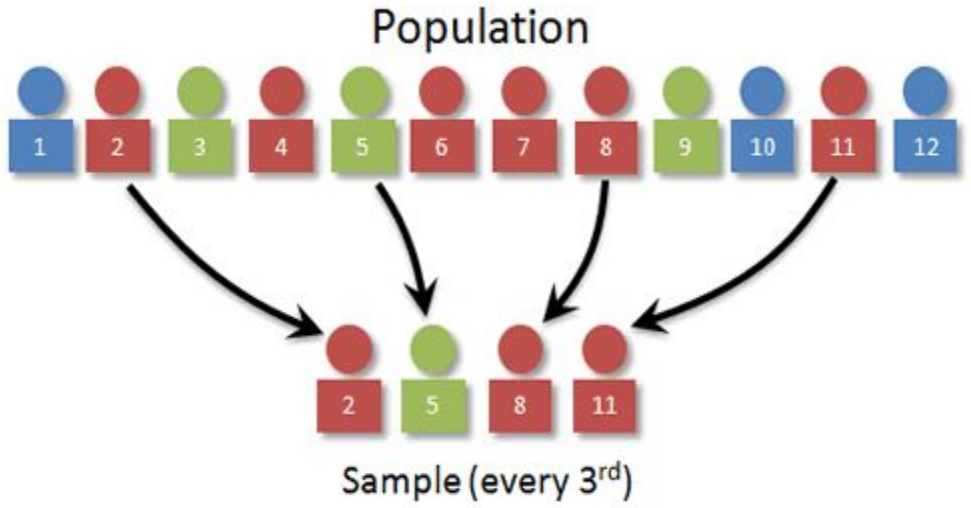
Pros: The main advantage of using systematic sampling over simple random sampling is its simplicity

Cons: The possible weakness of the method that may compromise the randomness of the sample is an inherent periodicity of the list.

Types of Probability Sampling

Systematic sampling works well when the individuals are already lined up in order. In the past, students have often used this method when asked to survey a random sample of ECC students. Since we don't have access to the complete list, just stand at a corner and pick every 10th* person walking by.

* Of course, choosing 10 here is just an example. It would depend on the number of students typically passing by that spot and what sample size was needed.



Types of Probability Sampling

Cluster Sampling

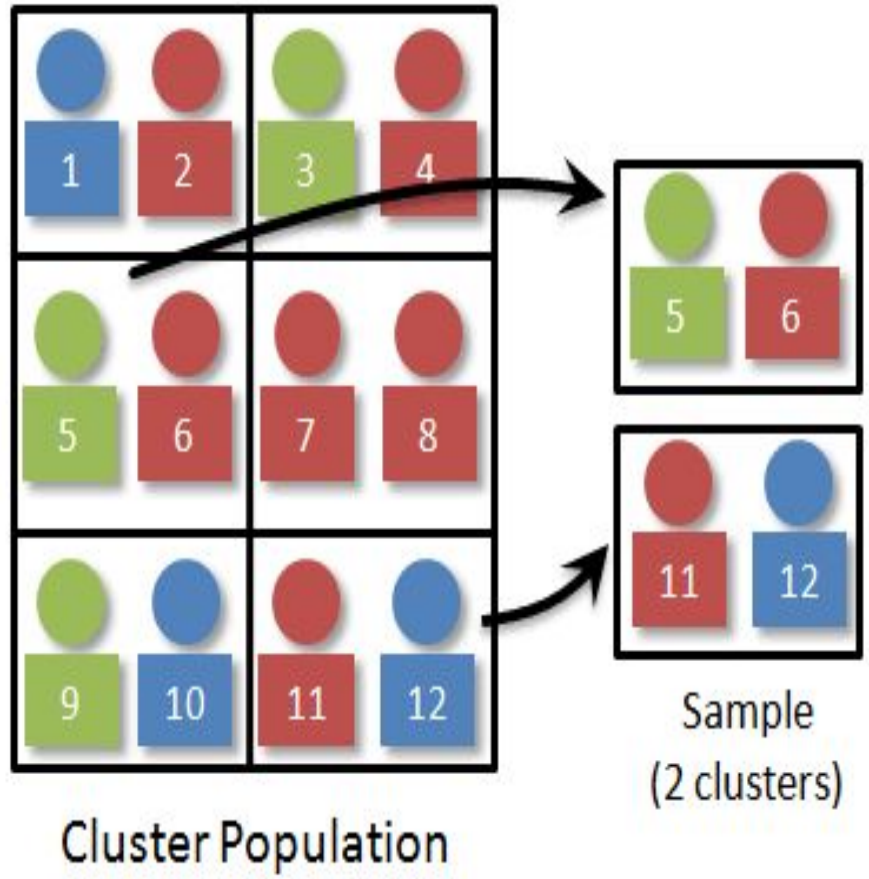
- **When:** we use cluster sampling when our population is already broken up into groups (*clusters*), and each cluster represents the population. That way, we just select a certain number of clusters.

- Cluster sampling is often confused with stratified sampling, because they both involve "groups". In reality, they're very different. In stratified sampling, we split the population into groups (strata) based on some characteristic.

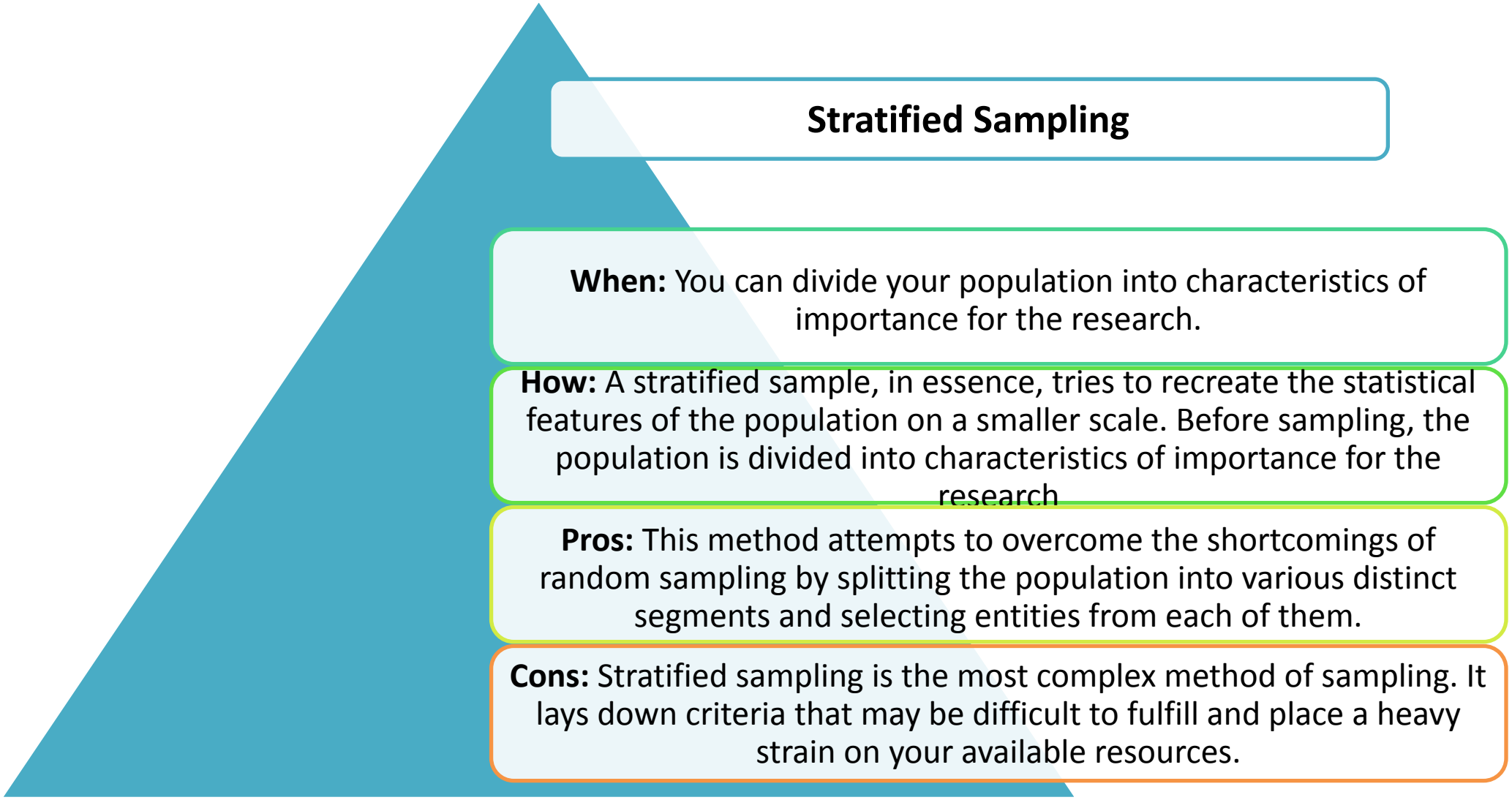
Types of Probability Sampling

One situation where cluster sampling would apply might be in manufacturing. Suppose your company makes light bulbs, and you'd like to test the effectiveness of the packaging. You don't have a complete list, so simple random sampling doesn't apply, and the bulbs are already in boxes, so you can't order them to use systematic. And all the bulbs are essentially the same, so there aren't any characteristics with which to stratify them.

To use cluster sampling, a quality control inspector might select a certain number of entire boxes of bulbs and test each bulb within those boxes. In this case, the boxes are the *clusters*.

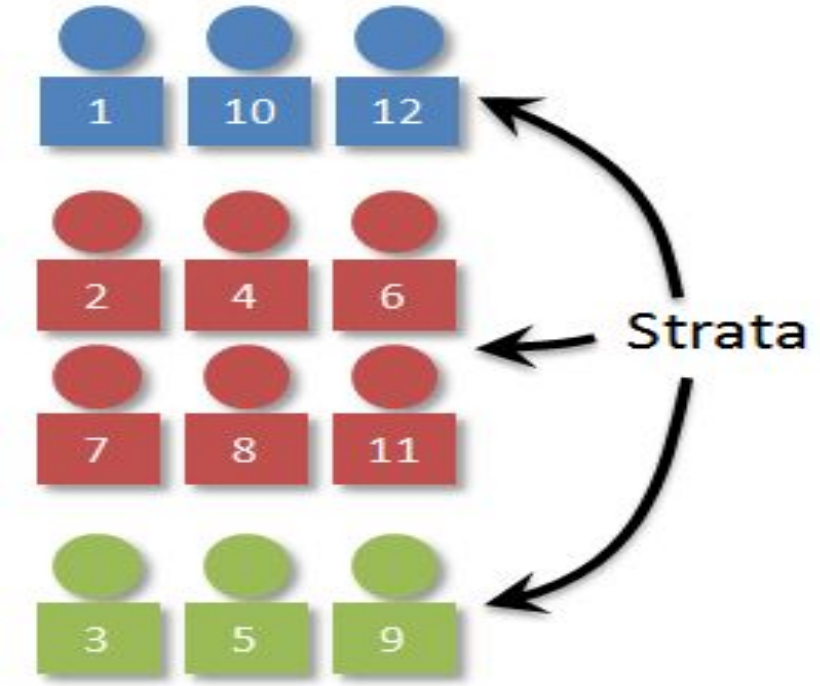


Types of Probability Sampling



Types of Probability Sampling

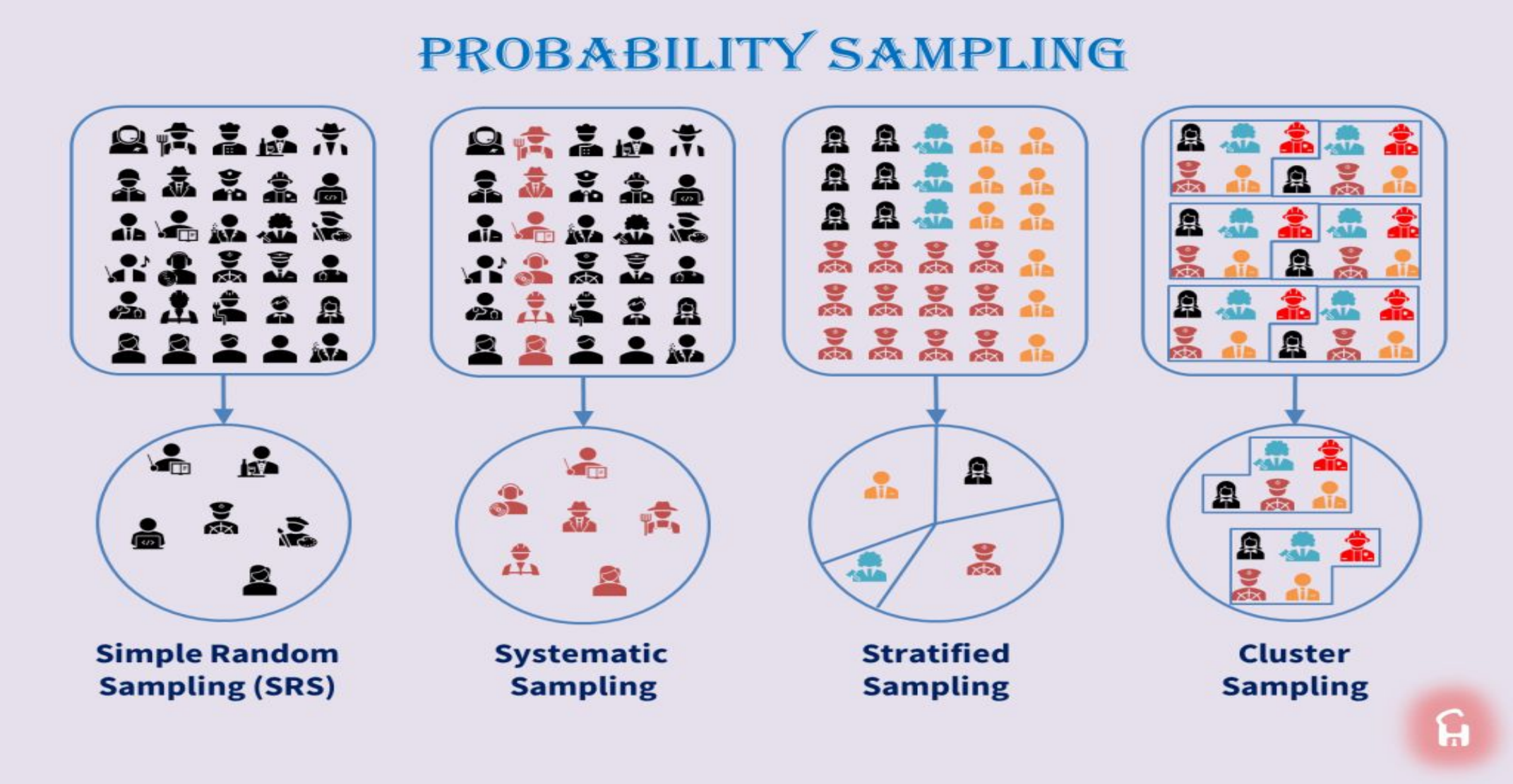
One easy example using a stratified technique would be a sampling of people at PCET. To make sure that a sufficient number of students, faculty, and staff are selected, we would stratify all individuals by their status - students, faculty, or staff. (These are the *strata*.) Then, a proportional number of individuals would be selected from each group.



Difference between Stratified & Cluster

Key	Stratified Sampling	Cluster Sampling
Meaning	Stratified sampling is one, in which the population is divided into homogeneous segments, and then the sample is randomly taken from the segments.	Cluster sampling refers to a sampling method wherein the members of the population are selected at random, from naturally occurring groups called 'cluster'.
Sample	Randomly selected individuals are taken from all the strata.	All the individuals are taken from randomly selected clusters.
Selection of population elements	Individually	Collectively
Homogeneity	Within group	Between groups
Heterogeneity	Between groups	Within group
Bifurcation	Imposed by the researcher	Naturally occurring groups
Objective	To increase precision and representation.	To reduce cost and improve efficiency.

Types of Probability Sampling



Uses of probability sampling

There are multiple uses of probability sampling:

- **Reduce Sample Bias:** Using the probability sampling method, the bias in the sample derived from a population is negligible to non-existent. The selection of the sample mainly depicts the understanding and the inference of the researcher. Probability sampling leads to higher quality data collection as the sample appropriately represents the population.
- **Diverse Population:** When the population is vast and diverse, it is essential to have adequate representation so that the data is not skewed towards one demographic. For example, if Square would like to understand the people that could make their point-of-sale devices, a survey conducted from a sample of people across the US from different industries and socio-economic backgrounds helps.
- **Create an Accurate Sample:** Probability sampling helps the researchers plan and create an accurate sample. This helps to obtain well-defined data.

Types of Non-Probability Sampling

Convenience Sampling

- **When:** During preliminary research efforts.

- **How:** As the name suggests, the elements of such a sample are picked only on the basis of convenience in terms of availability, reach and accessibility.

- **Pros:** The sample is created quickly without adding any additional burden on the available resources.

Types of Non-Probability Sampling

- **Quota Sampling**

- **When:** When you can characterize the population based on certain desired features.
- **How:** Quota sampling is the non-probability equivalent of stratified sampling that we discussed earlier. It starts with characterizing the population based on certain desired features and assigns a quota to each subset of the population.
- **Pros:** This process can be extended to cover several characteristics and varying degrees of complexity.

Types of Non-Probability Sampling

Snowball Sampling

- **When:** When you can rely on your initial respondents to refer you to the next respondents.
- **How:** Just as the snowball rolls and gathers mass, the sample constructed in this way will grow in size as you move through the process of conducting a survey. In this technique, you rely on your initial respondents to refer you to the next respondents whom you may connect with for the purpose of your survey.
- **Pros:** The costs associated with this method are significantly lower, and you will end up with a sample that is very

Sample Size

A **sample** is a selection of respondents chosen in such a way that they represent the total population as good as possible. However, instantly a new question comes to the forefront: *“How many people should my sample consist of?”*. Using a **correct survey sample size** is crucial for your research. After all, a sample that is too big will lead to the waste of precious resources such as time and money, while a sample that is too small will not allow you to gain reliable insights.

There are **two measures** that affect the accurateness of the data.

Required Sample Size†

Population Size	Confidence = 95% Margin of Error				Confidence = 99% Margin of Error			
	5.0%	3.5%	2.5%	1.0%	5.0%	3.5%	2.5%	1.0%
10	10	10	10	10	10	10	10	10
20	19	20	20	20	19	20	20	20
30	28	29	29	30	29	29	30	30
50	44	47	48	50	47	48	49	50
75	63	69	72	74	67	71	73	75
100	80	89	94	99	87	93	96	99
150	108	126	137	148	122	135	142	149
200	132	160	177	196	154	174	186	198
250	152	190	215	244	182	211	229	246
300	169	217	251	291	207	246	270	295
400	196	265	318	384	250	309	348	391
500	217	306	377	475	285	365	421	485
600	234	340	432	565	315	416	490	579
700	248	370	481	653	341	462	554	672
800	260	396	526	739	363	503	615	763
1,000	278	440	606	906	399	575	727	943
1,200	291	474	674	1067	427	636	827	1119
1,500	306	515	759	1297	460	712	959	1376
2,000	322	563	869	1655	498	808	1141	1785
2,500	333	597	952	1984	524	879	1288	2173
3,500	346	641	1068	2565	558	977	1510	2890
5,000	357	678	1176	3288	586	1066	1734	3842
7,500	365	710	1275	4211	610	1147	1960	5165
10,000	370	727	1332	4899	622	1193	2098	6239
25,000	378	760	1448	6939	646	1285	2399	9972
50,000	381	772	1491	8056	655	1318	2520	12455
75,000	382	776	1506	8514	658	1330	2563	13583
100,000	383	778	1513	8762	659	1336	2585	14227
250,000	384	782	1527	9248	662	1347	2626	15555
500,000	384	783	1532	9423	663	1350	2640	16055
1,000,000	384	783	1534	9512	663	1352	2647	16317
2,500,000	384	784	1536	9567	663	1353	2651	16478
10,000,000	384	784	1536	9594	663	1354	2653	16560
100,000,000	384	784	1537	9603	663	1354	2654	16584
300,000,000	384	784	1537	9603	663	1354	2654	16586

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Motivation for Hypotheses Testing

- Business: Will an investment in a mutual fund yield annual returns greater than desired value? (based on past performance of the fund)
- Medical: Is the incidence of diabetes greater among males than females?
- Social: Are women more likely to change mobile service provider than men?
- Engineering: Has the efficiency of the pump (η) decreased from its original value due to aging?

Hypotheses Testing

- The hypotheses is generally converted to a test of the mean or variance parameter of a population (or differences in means or variances of populations)
- A hypothesis a statement or postulate about the parameters of a distribution (or model)
  - Null hypothesis H_0 : The default or *status quo* postulate that we wish to reject if the sample set provides sufficient evidence (eg. $\eta = \eta_0$)
  - Alternative hypothesis H_1 : The alternative postulate that is accepted if the null hypothesis is rejected (eg. $\eta < \eta_0$)

Hypotheses Testing Procedure

- No hypotheses test is perfect. There are inherent errors since it is based on observations which are random
- The performance of a hypotheses test depends on
  - Extent of variability in data
  - Number of observations (Sample size)
  - Test statistic,(function of observations)
  - Test criterion (threshold)

Hypothesis Testing

Hypothesis testing is to test some hypothesis about parent population from which the sample is drawn.

- A hypothesis is a statement to be tested about the true value of population parameter using sample statistics.
- To test the validity of the claim or assumption about the population parameter.

1. A sample is drawn from the population and analyzed.
2. The results of the analysis are used to decide whether the claim is true or not.

Hypothesis Testing

- **Null Hypothesis (H0):** The hypothesis initial assumed to be true, although it may be either true or false on the sample data.
- 'Zero' implies 'no difference' between sample statistics and the parameter value.
- Hypothesis testing requires H0 considered to be true until it is proved to be false on the basis of results observed from the sample data.
- H0=μ $(\leq, =, \geq)\mu_0$

  where μ is population mean

  μ_0 is hypothesized value of μ

Hypothesis Testing

- **Alternative Hypothesis (H1):** H1 is true if H0 is rejected
- H1=$\mu \neq \mu_0$

where μ is population mean

μ_0 is hypothesized value of μ

- The hypothesis initial assumed to be true, although it may be either true or false on the sample data.
- 'one' implies 'difference' between sample statistics and the parameter value.
- **Level of Significance(α):** The likehood of rejecting a null hypothesis when it is true.

Errors in Hypotheses Testing

- Two Types of errors (Type I and Type II)

Decision → Truth ↓	H_0 is not rejected	H_0 is rejected
H_0 is true	Correct Decision $Pr = 1 - \alpha$	Type I error $Pr = \alpha$
H_1 is true	Type II error $Pr = \beta$	Correct Decision $Pr = 1 - \beta$

  - Typically the Type 1 error probability α (also called as level of significance of the test) is controlled by choosing the criterion from the distribution of the test statistic under the null hypothesis

Hypothesis Test

- **Significance level-** Probability of wrong decision that is p

p= Probability of occurrence (Wrong decision)

q= Probability of Non-occurrence (Correct decision)

p	Significance Level	Confidence Level
0.1	10%	90%
0.05	5%	95%
0.01	1%	99%
0.001	0.1%	99.9%

Hypothesis Test

p	Significance Level	Confidence Level	Two Tailed Test (Critical z Value)	One tailed Test (Critical z Value)
0.1	10%	90%	1.65	1.28
0.05	5%	95%	1.96	1.64
0.01	1%	99%	2.58	2.33
0.001	0.1%	99.9%	3.29	3.10

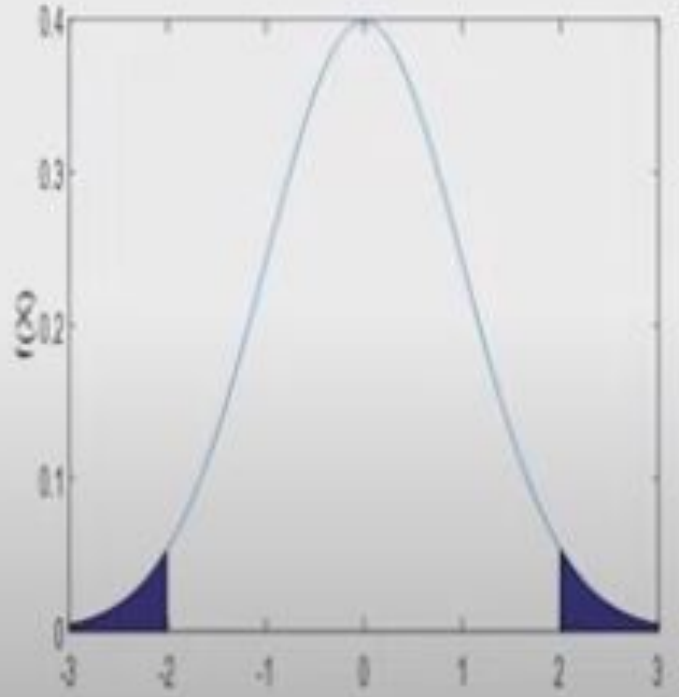
Two-sided and one-sided tests

- Two sided test,

$$H_0 : \mu = 0$$

$$H_1 : \mu \neq 0$$

- Test statistic standard normal RV z



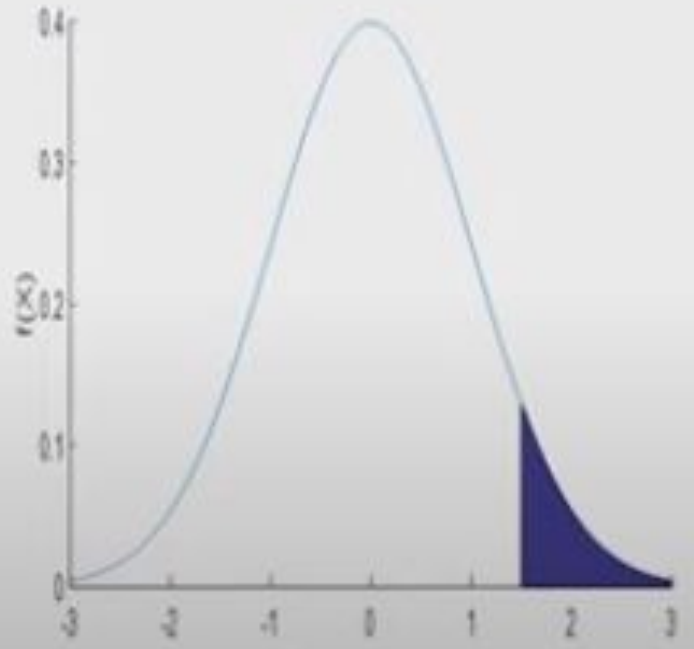
- Reject H_0 if $z \leq -2$ or $z \geq 2$

- One sided test

$$H_0 : \mu = 0$$

$$H_1 : \mu > 0$$

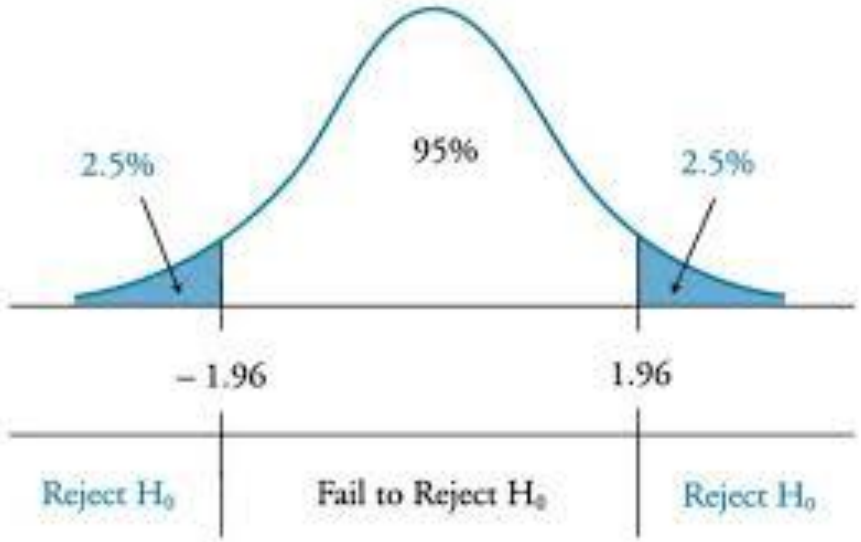
- Test statistic standard normal RV z



- Reject H_0 if $z \geq 1.5$

Hypothesis Test

Gender	Group	Weight(Kg)	Hight (m)
M	Adult	55	1.2
F	Child	30	1.1
F	Child	26	1
M	Elderly	68	1.4
M	Adult	72	1.5
F	Elderly	58	13



$P \leq 0.05$ **Significant Level** α

Hypothesis Test

Prob. 1: If we want to find proportion of one categorical

Gender	Group	Weight(Kg)	Hight (m)
M	Adult	55	1.2
F	Child	30	1.1
F	Child	26	1
M	Elderly	68	1.4
M	Adult	72	1.5
F	Elderly	58	13

	One Categorical
	There is no difference
	There is difference
Test	**One sample proportion Test**

$$P \leq 0.05$$

Hypothesis Test

Prob. 2: If we want to analyze proportion of two categorical

Gender	Group	Weight(Kg)	Hight (m)
M	Adult	55	1.2
F	Child	30	1.1
F	Child	26	1
M	Elderly	68	1.4
M	Adult	72	1.5
F	Elderly	58	13

	Two Categorical
	There is no difference
	There is difference
Test	**Chi Square Test**

$$P \leq 0.05$$

Hypothesis Test

Prob. 3: If we want to analyze proportion of one numerical variable

Gender	Group	Weight(Kg)	Hight (m)
M	Adult	55	1.2
F	Child	30	1.1
F	Child	26	1
M	Elderly	68	1.4
M	Adult	72	1.5
F	Elderly	58	13

	One Numerical Variable
	There is no difference
	There is difference
Test	**t-Test**

$$P \leq 0.05$$

Thank you

Feel free to ask your questions